

Ansys Fluent Simulation Settings & Benchmark Guide

Transregio SFB-TR40 · Test Case 1: Single-Element GCH₄/GOX Chamber

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1 Executive Summary & TC1 Benchmark Overview

This reference document tracks the physical modeling, numerical settings, meshing rules, and output requirements for the **Transregio SFB-TR40 Test Case 1 (TC1)** benchmark. The experimental campaign was conducted at the Lehrstuhl für Turbomaschinen und Flugantriebe (LFA) of the Technische Universität München (TUM) to provide high-fidelity validation data for numerical combustion models of gaseous oxygen (GOX) and gaseous methane (GCH₄) rocket thrust chambers.

1.1 Chamber Architecture & Injector Geometry

The chamber features a square bore with a single coaxial shear injector and a rectangular converging-diverging nozzle throat:

- **Chamber Bore:** Square cross section of 12×12 mm, composed of two segments: Segment 1 (174 mm) and Segment 2 (145 mm), yielding 319 mm of constant duct.
- **Nozzle Section:** 20 mm converging section tapering to a rectangular throat (4.8 mm height $\times$ 12.0 mm width), yielding a contraction ratio $\epsilon = 2.5$ and chamber Mach number $Ma \approx 0.24$. The length from faceplate to throat is 303 mm; total physical assembly length is 339 mm.
- **Coaxial Shear Injector:** Flush mounting with injection face (recess = 0 mm). Central GOX core of $\varnothing 4.0$ mm inner diameter with 0.5 mm post wall thickness. Surrounding GCH₄ annulus of $\varnothing 5.0$ mm ID $\times$ $\varnothing 6.0$ mm OD.
- **Torch Igniter:** Side wall mounted at $z \approx 130$ mm from faceplate. Not geometrically modeled in CFD; ignition is numerically initiated via high-temperature/product patching.
- **Thermal Construction:** Capacitively cooled oxygen-free copper (Cu-HCP: $\rho = 8940$ kg/m³, $k = 385$ W/(m · K)). Solved as a fixed non-uniform wall temperature problem, not conjugate heat transfer.

1.2 Nominal Operating Conditions (Test TFC-R-20-26-30-10)

The simulation corresponds to the hot-run evaluation window centered at $t_1 = 2.01$ s averaged over 0.5 s of steady burn:

Critical Coordinate Origin Note (Z_OFFSET_M)

In the experimental Thermtest data (Table 9 of benchmark specification), the axial coordinate $z^{**} = 0$ is defined at the start of the first copper segment, which is located **29 mm upstream of the injector faceplate**. When using a CAD/mesh where the faceplate is $z = 0$, an axial

Parameter	Full Domain Value	Quarter-Symmetry (÷4)
Chamber Pressure (P_c)	20.0 bar nominal	20.0 bar
Mixture Ratio (O/F)	2.62	2.62
Total Mass Flow Rate ($\dot{m}_{\text{tot}}$)	0.062 kg/s	0.01550 kg/s
Oxygen Mass Flow ($\dot{m}_{\text{Ox}}$)	0.045 kg/s (feed: 26.4 bar, 278 K)	0.01125 kg/s
Methane Mass Flow ($\dot{m}_{\text{CH}_4}$)	0.017 kg/s (feed: 24.4 bar, 269 K)	0.00425 kg/s
Porous Plate Pressure Drop	$\Delta P/P \approx 9\%$ (Ox), 11% (CH ₄)	Accounts for inlet manifold drop

offset of $\Delta z = 0.029$ m must be applied:

$$z_{\text{Thermtest}} = z_{\text{mesh}} + 0.029 \text{ m}$$

This explains the parameter `Z.OFFSET_M = 0.029` defined in the boundary UDFs.

2 Mandatory Deliverables (§6 Benchmark Specification)

The benchmark specification explicitly requires five primary outputs for validation against experimental data:

1. **Axial Wall Pressure Distribution $p(z)$:** Profile along the side wall compared against the 9 calibrated piezoresistive transducers (PC0 through PC8):

Transducer	PC0	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
z^{**} [mm]	0.5	34.5	68.5	102.5	136.5	170.5	204.5	238.5	272.5

2. **Wall Heat Flux Profiles $\dot{q}_w(z)$:** Evaluated along the two prescribed thermocouple lines:
 - Upper wall centerline: T_{up} line ($x = 0, y = +6$ mm).
 - Side wall centerline: T_{side} line ($x = +6$ mm, $y = 0$).
3. **2D Iso-Plots & Planar Contours:** Mid-plane slices ($y = 0$ plane) displaying temperature, velocity magnitude, and major species fields.
4. **Axial Product Mass Fraction Evolution:** Centerline profiles of major species (H₂O, CO₂, CO, H₂, O₂, CH₄).
5. **Centerline OH Radical Profile:** Mass/mole fraction of OH along the chamber centerline, serving as the primary physical marker of the flame anchoring zone and reaction front length.

3 Computational Budget, Meshing Strategy & Sizing Physics

3.1 Project Phase Breakdown & Time Estimates

3.2 Solver Runtime Scaling & ETA Diagnostics (4 CPU Cores, 192,688 Nodes)

The current optimized mesh contains **192,688 nodes** (~150,000 cells), well below the Ansys Student 1M cap.

Project Phase	Scope of Work	Estimated Time
Geometry & CAD	SpaceClaim fluid domain extraction, symmetry cuts, VOR bodies	4–8 hours
Meshing & Sizing	Inflation layer setup, shear layer VOR, mesh quality checks	4–8 hours
Solver Configuration	Mechanism import, species properties, boundary UDFs, monitors	4–8 hours
CFD Calculation	Density-based coupled solving, flame stabilization, convergence	1–2 days (wall clock)
Post-Processing	Profile extraction (PC0–PC8, T_{up} , T_{side}), OH plots, reports	4–6 hours
Total Active Effort	Hands-on modeling and verification	~15–25 hours

State / Model	Time / Iteration	Iterations	Estimated Wall-Clock
Current Mesh (192k nodes, FR/ED stable)	~ 0.8–1.5 s / iter	500 (batch)	~ 8–12 minutes
Full Convergence (192k nodes, 16 sp)	~1.0–1.5 s / iter	3,000–5,000 iters	~ 45–75 minutes
EDC (Stiff ODE, 16 sp)	~10–20 s / iter	4,000 iters	~ 12–20 hours

Why Fluent Displayed an Initial 44-Hour ETA (and Why It Drops)

At iteration 10, Fluent reported an ETA of 44 hours (down from 203 hours at iteration 2). This initial inflation is caused by two temporary factors:

1. **Divergence Back-off Retries:** In each startup iteration, Fluent encountered divergence checks, triggering 3–4 sub-iterations stepping down from 10^{-6} s to 2.47×10^{-8} s. Each iteration effectively solved the coupled AMG equations four times.
2. **Disk I/O from 4 Animations Every Iteration:** Fluent was writing four graphic animation sequence files (.cxa) at every single iteration, adding multi-second graphics overhead.

Once the flame anchors and the time step stabilizes without divergence retries, the iteration time drops to ~1.0 second per iteration, completing 500 iterations in **under 12 minutes**.

3.3 Bodies of Influence (VOR) & Mesh Sizing

To maintain mesh quality and resolve the shear layer without cell count explosion, four Body of Influence (BOI / VOR) volumes are defined in CAD:

SpaceClaim / Meshing Rule for VORs

In the CAD tree, **Suppress** all four VOR bodies (`inletVOR`, `shearLayerVOR`, etc.). Suppressed bodies remain selectable as geometry scopes for *Mesh* → *Insert* → *Sizing* → *Body of Influence*, but will **not** be meshed as separate fluid volumes or exported to Fluent.

Region / Body	Element Size	Physical Purpose
inletVOR	0.60 mm	Resolves GOX post lip (0.5 mm wall) and methane annular jet
shearLayerVOR	0.60 mm	Captures turbulent mixing layer and flame anchoring zone
convergentVOR	2 mm	Resolves accelerating flow into nozzle contraction
nozzleVOR	1.2 mm	Resolves transonic acceleration and sonic throat shock structure
Chamber Bulk Duct	3.00 mm	Coarse structured/swept hex mesh away from core shear layer

3.4 Near-Wall Resolution: Smooth Transition Inflation (Max 8 Layers)

While theoretical viscous sublayer resolution ($y^+ \approx 1$) yields an initial cell height $y_1 \approx 0.15 \mu\text{m}$, directly prescribing $0.15 \mu\text{m}$ onto a coarser chamber duct creates extreme aspect ratios ($> 10^4$) and high cell skewness, severely destabilizing the density-based coupled solver.

Adopted Setup: Smooth Transition, Maximum 8 Layers

- **Transition Scheme:** Smooth Transition with a transition ratio of 0.272 and growth rate of 1.20.
- **Maximum Layers:** 8 layers, maintaining healthy cell aspect ratios (< 100) adjacent to duct faces.
- **Resulting Mesh Size:** Total node count stands at **192,688 nodes** ($\sim 150,000$ cells), providing a robust balance between wall gradient capture and numerical stability.
- **Turbulence Wall Treatment:** SST k - ω automatically blends from resolved to analytical sublayer profiles across cells where $1 < y^+ < 10$.

4 Combustion Chemistry & Reaction Mechanism Selection

4.1 GRI-Mech 3.0 Memory Failure Rationale

Full GRI-Mech 3.0 contains 53 species and 325 elementary reactions. In Ansys Fluent's density-based coupled solver, each cell solves a coupled Algebraic Multigrid (AMG) system across all transported variables simultaneously:

$$N_{\text{coupled}} = 5 \text{ (continuity, 3-momentum, energy)} + 2 \text{ (SST } k\text{-}\omega) + (N_{\text{species}} - 1) \approx 59 \text{ equations/cell}$$

Memory allocation scales quadratically with equation count ($O(N_{\text{coupled}}^2)$). On a 1M cell mesh, GRI-Mech 3.0 requires > 25 GB of RAM simply to build the AMG matrix during hybrid initialization, causing immediate allocation failures on typical workstation hardware.

4.2 Selection of 16-Species Reduced Mechanism

- **Why Built-in Global Mechanisms Fail:** Fluent's default 2-step and 4-step methane mechanisms track only 5 species (CH_4 , O_2 , CO_2 , H_2O , N_2). They do *not* track the OH radical, making the mandatory §6 benchmark deliverable impossible to extract.
- **Chosen Mechanism (gri16_reduced.inp):** A 16-species skeletal mechanism (60 reactions) retaining essential radicals:

H₂, H, O, O₂, OH, H₂O, HO₂, H₂O₂, CH₃, CH₄, CO, CO₂, HCO, CH₂O, N₂, AR

- **Inert Stripping:** Since propellants are pure GOX and pure GCH₄, N₂ and AR can be deactivated from transport to save memory.
- **Alternative Non-Premixed (Flamelet) Model:** If runtime memory is constrained, the Non-Premixed Combustion model solves only 2 transport equations (mixture fraction f and variance f'^2), retrieving OH from an offline pre-tabulated chemistry library.

5 Ansys Fluent Detailed Simulation Settings

5.1 General & Physical Models

Parameter	Selected Setting / Value
Solver Type	Density-Based, Steady state
Velocity Formulation	Absolute
2D / 3D Space	3D (or Quarter-Symmetry 3D)
Energy Equation	On
Viscous Model	SST k - ω (Shear Stress Transport)
Viscous Options	Low-Re Corrections enabled ($y^+ \approx 1$ mesh); default blending constants
Species Model	Species Transport
Reactions	Volumetric enabled
Combustion Model	Finite-Rate / Eddy-Dissipation (Turbulence-Chemistry interaction)
Model Constants	$A = 4.0$, $B = 0.5$ (standard Magnussen constants)

5.2 Materials & Chemistry Properties (chemkin-import)

Parameter	Selected Setting / Value
Mixture Material	chemkin-import (File: gri16_reduced (1).inp, 16 species, 43 reactions)
Species List	H ₂ , H, O, O ₂ , OH, H ₂ O, HO ₂ , H ₂ O ₂ , CH ₃ , CH ₄ , CO, CO ₂ , HCO, CH ₂ O, N ₂ , AR
Density Method	ideal-gas (mandatory for compressible density-based solver)
c_p (Specific Heat)	mixing-law
Thermal Conductivity	mass-weighted-mixing-law
Dynamic Viscosity	mass-weighted-mixing-law
Mass Diffusivity	constant-dilute-approx

5.3 Cell Zone Conditions (Fluid Domain)

Parameter	Selected Setting / Value
Zone Name	fluid
Material Name	chemkin-import
Reaction	Volumetric reactions enabled with imported mechanism
Laminar Flame Speed	Disabled (diffusion/shear flame)

5.4 Boundary Conditions

1. Oxidizer Inlet (gox_inlet) – Mass-Flow Inlet

Parameter	Selected Setting / Value
Mass Flow Rate	0.045 kg/s (Full domain)
Total Temperature	278.0 K
Direction Specification	Normal to Boundary
Species Mass Fraction	O ₂ = 1.0, all other species = 0.0
Turbulence Specification	Intensity 5%, Viscosity Ratio 10%

2. Fuel Inlet (gch4_inlet) – Mass-Flow Inlet

Parameter	Selected Setting / Value
Mass Flow Rate	0.017 kg/s (Full domain)
Total Temperature	269.0 K
Direction Specification	Normal to Boundary
Species Mass Fraction	CH ₄ = 1.0, all other species = 0.0
Turbulence Specification	Intensity 5%, Viscosity Ratio 10%

3. Nozzle Exit (outlet) – Pressure Outlet

Parameter	Selected Setting / Value
Gauge Static Pressure	1.0 bar (Exit flow chokes upstream at the 4.8×12 mm throat; exit back-pressure has negligible upstream influence)
Backflow Total Temperature	300.0 K
Backflow Species	Products (CO ₂ = 0.1, H ₂ O = 0.1)

4. Upper/Lower Chamber Walls (wall_top_bottom) – Thin Wall

Parameter	Selected Setting / Value
Wall Material	Aluminum (active setup; copper Cu-HCP in physical hardware)
Thermal Boundary Condition	Temperature (Dirichlet condition; interface T enforced via UDF)
Temperature Formulation	User-Defined Function: <code>udf_chamber_top-T</code>
Wall Motion / Shear	Stationary Wall, No-Slip condition
Wall Thickness / Heat Gen	Thickness = 0.0 mm, Heat Generation = 0.0 W/m ³

5. Side Chamber Walls (wall_sides) – Thick Wall

Parameter	Selected Setting / Value
Thermal Boundary Condition	Temperature
Temperature Formulation	User-Defined Function: <code>udf_chamber_side-T</code>
Wall Motion / Shear	Stationary Wall, No-Slip condition
Wall Thickness / Heat Gen	Thickness = 0.0 mm, Heat Generation = 0.0 W/m ³

6. Wall Temperature UDF (chamber_wall_temperatures.c)

Parameter	Selected Setting / Value
Source File	<code>chamber_wall_temperatures.c</code> (Compiled UDF library)
Profiles Exported	<code>chamber_top-T</code> (Upper wall) and <code>chamber_side-T</code> (Side wall)
Z_OFFSET_M Parameter	0.029 m (for faceplate origin) 0.0 m (if origin at segment 1)
Interpolation Rule	Linear spline through Thermtest 10 mm measurement coordinates

5.5 Surfaces & Monitoring Probes

Surface Name	Type & Geometric Coordinates
pc0 through pc8	Point surfaces located on side wall ($x \approx 6.0$ mm, $y \approx 0.0$ mm): $z = 0.5, 34.5, 68.5, 102.5, 136.5, 170.5, 204.5, 238.5, 272.5$ mm
centerline_till_throat	Line / rake from injector faceplate (0,0,0) to throat (0,0,303 mm) along axis
plane_y0	Planar surface at $y = 0$ for mid-plane contour visualization

5.6 Cell Registers & Ignition Patching

Parameter	Selected Setting / Value
Register Name	injector
Geometric Shape	Sphere
Sphere Center	($x = 0.0$ m, $y = 0.0$ m, $z = 0.005$ m)
Sphere Radius	0.005 m (5.0 mm, enveloping the near-injector shear layer)
Patch Temperature	2500 K
Patch Species	H ₂ O ≈ 0.20 , CO ₂ ≈ 0.15 , OH ≈ 0.01
Patch Sequence	Mandatory immediately after Hybrid Init, before Calculate

Why Product Species Patching is Mandatory for Combustion

In the Finite-Rate / Eddy-Dissipation combustion model, the turbulent combustion rate is strictly limited by the product concentration:

$$R_{ED} = A \cdot \rho \frac{\varepsilon}{k} \min \left([Reactants], B \sum \frac{Y_{Products}}{M_{w,Products}} \right)$$

If product species are **not patched** ($Y_{Products} = 0$), the Eddy-Dissipation reaction rate is **identically zero**. Furthermore, because the feed streams enter cold (CH₄ at 269 K, O₂ at 278 K), finite-rate autoignition does not occur without initial radical carriers. The flame will **never ignite**, and the hot temperature patch simply blows out or diffuses away, causing unburned shear-layer divergence.

5.7 Solution Numerical Methods

Parameter	Selected Setting / Value
Formulation	Implicit
Flux Type	AUSM (switched from Roe-FDS after persistent pressure clipping and divergence retries at startup)
Gradient Spatial Scheme	Least Squares Cell Based
Flow	First Order Upwind
Turbulent Kinetic Energy	First Order Upwind
Specific Dissipation Rate	First Order Upwind
Species Transport Equations	First Order Upwind
Pseudo Time Method	Global Time Step
Warped-Face Gradient Correction	On (Enable checkbox to stabilize high-aspect inflation cells)
Solution Limits (Controls)	$P_{\min} = 10,000 \text{ Pa}$, $P_{\max} = 5 \times 10^6 \text{ Pa}$, $T_{\min} = 250 \text{ K}$, $T_{\max} = 3500 \text{ K}$
High Order Term Relaxation	Off during 1st order; On when switching to 2nd order

5.8 Run Calculation Parameters

Parameter	Selected Setting / Value
Time Scale Factor	0.02
Time Step Method	Automatic
Length Scale Method	Conservative
Number of Iterations	150
Reporting Interval	10 iteration
Reporting Interval (Graphs)	1 iteration
Profile Update Interval	1 iteration
Solution Steering	Off

5.9 Report Definitions & Convergence Monitors

Monitor Name	Definition & Purpose
cfl-number	Courant-Friedrichs-Lewy monitor to track time-stepping stability
total_heat_flux_top_bottom	Surface Integral → Total Surface Heat Flux on wall_top_bottom
total_heat_flux_side	Surface Integral → Total Surface Heat Flux on wall_sides
static_pressure_at_left_wall	Surface Facet Average → Static Pressure monitor
oh_centerline_monitor	Vertex Average → Mass fraction of OH along centerline_till_throat
mass_flow_balance	Net flux monitor ($\dot{m}_{\text{in}} - \dot{m}_{\text{out}}$), settles near 0.0 kg/s at convergence

5.10 Calculation Activities & Solution Animations

Activity	Execution Frequency / Details
Autosave Case & Data	Write every 200–500 iterations with automatic incrementing filename
Centerline Animations	Record every 10–25 iterations (writing every 1 iter throttles I/O)
Animation Objects	o2-center, velo-center, temp-center, ch4-center on mid-plane
Execute Commands	Unused / Default
Cell Register Operations	Unused / Default

6 Action Checklist & Verification Protocol

Use the checklist below to verify settings during simulation setup and post-processing. Settings can be adapted or commented out as needed:

Pre-Run Verification Checklist

- ✓ **Mechanism Validation:** Confirm Number of Volumetric Species = 16 in the Materials panel. Ensure OH is included.
- ✓ **Quarter-Domain Mass Flows:** If running quarter symmetry, verify mass flows are scaled ($\dot{m}_{\text{Ox}} = 0.01125 \text{ kg/s}$, $\dot{m}_{\text{CH}_4} = 0.00425 \text{ kg/s}$).
- ✓ **Symmetry Planes Tagged:** Ensure the cut planes ($X = 0$ and $Y = 0$) are set to symmetry, not default wall.
- ✓ **Ignition Patch Executed:** After Hybrid Initialization, immediately patch sphere injector with $T = 2500 \text{ K}$ and product species before pressing Calculate.
- ✓ **UDF Origin Offset Checked:** Verify that Z.OFFSET_M matches mesh coordinate origin (0.029 m if origin is at faceplate).

- ✓ **Discretization Ramp:** Begin iterations with 1st Order Upwind. Once flame anchors and residuals stabilize, switch Flow, Turbulence, and Species to 2nd Order Upwind with High Order Term Relaxation enabled.